

REVIEW OF THE PAST LESSON





Agronomist



Aquaculturist



Livestock Technician



Agribusiness Manager



Crop Production Technician

AGRICULTURAL TOOLS, IMPLEMENTS, EQUIPMENTS AND FARM HAZARDS



LESSON 2

LEARNING COMPETENCIES

- **The learners differentiate agricultural tools, implement and equipment; identify the different tools and equipment used in agriculture; discuss safety procedures in the farm operations when using tools, equipment and implements; and discuss hazards and risks in farm operations.**

LEARNING OBJECTIVES

- **Discuss agricultural tools, implements and equipment and their usage, safety and maintenance.**
- **Explain safety procedures in farm operations when using tools, equipment and implements; and**
- **Explain the ways on how to prevent hazards and risks in farm operations.**

OVERVIEW OF THE LESSON

- **The productivity, efficiency and overall crop output of modern farming techniques are due to the quality of agricultural tools, implement and equipment. Many tasks require the use of these instruments, which may vary from simple hand tools to highly developed machines.**

OVERVIEW OF THE LESSON

- **Tools, implements and equipment are necessary for farming activities. The output of agricultural crops is greatly impacted by them. Using these makes the task considerably quicker and easier when they are available, depending on the kind and extent of the operation that needs to be done on the farm.**

“FARM FRENZY” Pictionary

- 1. Form a 5 group each and choose who will represent the group.**
- 2. A secret name of tools, implements or equipment will be revealed to the representative, and he/she is given one minute to draw the item on the board without talking.**
- 3. The teammates have to guess the tool based on the drawing.**
- 4. Award points go to the team with the highest number of correct guesses.**

MAJOR DIFFERENCE BETWEEN FARM IMPLEMENTS AND EQUIPMENT

FARM TOOLS	FARM IMPLEMENTS	FARM EQUIPMENT
Typically, handheld	Larger, more complex than tools	Machine-powered
Perform specific tasks	Animal-powered, mechanized	Perform larger tasks
Small in size	For large-scale operations	Large in size
Easy to handle	For large-scale operations	Advanced technology
Perform detailed farm work (small-scale tasks)		Perform a range of tasks

FARM TOOLS

- **farm tools are simple, handheld devices used in agricultural activities. these tools are typically small, manually operated, and designed for tasks like planting, weeding and basic soil preparation.**

Tools

Uses

Safety

Maintenance



Spade – is a shovel with a flat blade used for digging, cutting, moving and cleaning up spills.

Use the spade for its intended purpose, such as digging, edging or turning soil. Position your hands correctly on the handle to maintain control and leverage.

Keep the blade sharp using a metal file or whetstone. After each use, wash of any dirt or debris and store it in a dry place to prevent rust.

Tools



Uses

Hand fork – is a tool with a pronged metal head and a wooden handle used for loosening, aerating compost piles to encourage decomposition and turning soil as well as for lifting and transplanting seedlings.

Safety

Avoid using the hand fork in extremely hard or rocky soil, which can cause the tool to slip or break

Maintenance

Keep the tines sharp using a metal file. Wash off any dirt or debris after each use and store it in a dry place to prevent rust.

Tools



Uses

Hoe– is a tool with a flat metal blade at right angles to a wooden handle; used for weeding , digging, hilling soil around crops and breaking up compacted soil for planting and creating furrows for planting seeds.

Safety

Be cautious of where you place the hand hoe to avoid tripping hazards. Do not use the hand hoe as a lever or prying tool, as this can cause damage or injury and avoid using the hand hoe in extremely hard or rocky soil, which can cause the tool to slip or break.

Maintenance

Keep the blade sharp using a metal file or whetstone. After each use, wash off any dirt or any debris and store it in a dry place to prevent rusting

Tools



Uses

Rake – is a garden tool with handle and a row of teeth used for raking leaves, grass clippings and other debris from lawns, gardens and walkways. It is also used for smoothing soil after planting seeds and spreading mulch or compost in a thin layer.

Safety

Ensure the tines are secure and the handle is free of splinters or cracks, and the handle is attached firmly to the rake head. Use a firm but controlled motion by holding the rake correctly with both hands to maintain control and leverage and to avoid slipping or losing grip.

Maintenance

Remove any debris stuck between the tines. Check the handle for cracks or loose connections and repair or replace as needed. Regularly check and straighten bent tines if necessary to maintain effectiveness and safety. Store it in a dry place.

Tools



Uses

Shovel – is a tool with a broad, deep blade and a short handle; used for scooping, lifting and moving loose material such as soil, sand or gravel, compost, manure or mulch spreading mulch or gravel and cleaning up spills. It is also used for digging holes, trenches and planting beds.

Safety

Do not use the shovel as a lever or prying tool, as this can cause damage or injury. Avoid using the shovel in extremely hard rocky soil, which can cause the tool to slip or break.

Maintenance

Keep the blade sharp using a metal file or whetstone. After each use, wash off any dirt or debris and store it in a dry place to prevent rust. Regularly check and sharpen the blade if necessary to maintain effectiveness and safety.

Tools



Uses

Axe – is a tool with a heavy head and a sharp blade mounted on a handle; used for wood chopping, dividing up logs, small tree removal (based on axe size), felling trees and splitting it, limbering branches, and scratching logs (for construction)

Safety

Clear the area of any tripping hazards and ensure you have enough space to swing the axe safely. Avoid using the axe in poor weather conditions, which can make surfaces slippery.

Maintenance

Sharpening: Keep the blade sharp using a whetstone or file. Aim to maintain the original bevel angle.
Cleaning: After each use wipe off dirt, sap and debris with a damp cloth. For tougher grime, use a wire brush and a light solvent (like mineral oil).
Rust Prevention: Apply a thin coat of oil (like linseed oil) on the metal blade to prevent rust. Wipe off any excess oil.

Tools



Uses

Axe – is a tool with a heavy head and a sharp blade mounted on a handle; used for wood chopping, dividing up logs, small tree removal (based on axe size), felling trees and splitting it, limbering branches, and scratching logs (for construction)

Safety

Clear the area of any tripping hazards and ensure you have enough space to swing the axe safely. Avoid using the axe in poor weather conditions, which can make surfaces slippery.

Maintenance

***Handle Care:* Regularly inspect the wooden handle for cracks, splinters or looseness. Sand smooths any rough spots. Oil the handle occasionally with the linseed oil to prevent drying and cracking. Store the axe in a dry, well-ventilated place, away from the direct sunlight. Ideally, hang it vertically to prevent warping.**

Tools



Uses

Sickle – is a hand-held harvesting tool with a curved blade used for gathering materials, such as tall grasses for various uses or reeds for basket weaving and in cutting grains and weeds.

Safety

Hold the sickle correctly with both hands to maintain control and leverage using a firm, controlled motion to avoid slipping or losing grip. Swing the sickle away from your body to reduce the risk of injury. Do not use the sickle as a hammer or prying a tool.

Maintenance

Make sure to clean the blade from any plant material, sap or dirt after each use. For milder dirt, use moist towel; for more stubborn dirt use a wire brush. Check for looseness, splinters or cracks in the handle. Sand smooth any ridges to improve grip and to keep the hardwood handle from drying out and cracking, occasionally coat it with linseed oil. Keep your sickle out of direct sunlight and in a dry, well-ventilated area hang it vertically to avoid warping.

QUESTION.....

A farmer has limited resources and can only bring three farm tools with him to a newly opened plot of land for vegetable production. If the land is uneven, filled with weeds, and has rocky soil, which three tools should he choose and why? Explain how these tools will complement each other in preparing the land efficiently.

LEARNERS' TAKEAWAYS

- **WHAT DID YOU LEARN ABOUT FARM TOOLS, THEIR USES, SAFETY PRECAUTIONS AND PROPER MAINTENANCE?**
- **HOW THESE TOOLS HELP OUR FARMERS**
- **WHAT DO YOU THINK WILL HAPPEN IF THESE FARM TOOLS DON'T EXIST? WHAT WILL BE THE IMPACT OF THIS ON CROP PRODUCTION?**

REFLECTION ON LEARNING

- **AFTER LEARNING ABOUT THE DIFFERENT FARM TOOLS, WHICH TOOL DO YOU THINK IS THE MOST IMPORTANT IN STARTING A VEGETABLE GARDEN AT HOME, AND WHY? HOW WILL THIS KNOWLEDGE HELP YOU IF YOU ARE TASKED TO HELP IN FARM OR GARDEN WORK IN THE FUTURE?**

FORMATIVE ASSESSMENT

1. WHICH FARM TOOL IS BEST USED FOR DIGGING SOIL AND MAKING PLANTING HOLES?

A. SPADE

B. RAKE

C. HOE

D. SICKLE

2. A FARMER WANTS TO LOOSEN THE SOIL AROUND VEGETABLE PLANTS WITHOUT DAMAGING THEIR ROOTS. WHICH TOOL SHOULD HE USE?

A. SHOVEL

B. HAND FORK

C. AXE

D. HOE

3. WHICH TOOL IS COMMONLY USED FOR LEVELING AND SMOOTHING THE SOIL AFTER PLOWING?

A. HOE

B. RAKE

C. SPADE

D. SICKLE

4. WHICH TOOL IS USED FOR CUTTING DOWN SMALL TREES AND CHOPPING WOOD FOR FARM USE?

A. SHOVEL

B. SICKLE

C. AXE

D. HOE

5. A FARMER NEEDS TO HARVEST RICE CROPS QUICKLY. WHICH TOOL SHOULD HE USE?

A. SPADE

B. SHOVEL

C. HOE

D. SICKLE

ANSWER KEY:

1. A

2. B

3. B

4. C

5. D

REVIEW OF THE PAST LESSON

GUESS WHO?



+

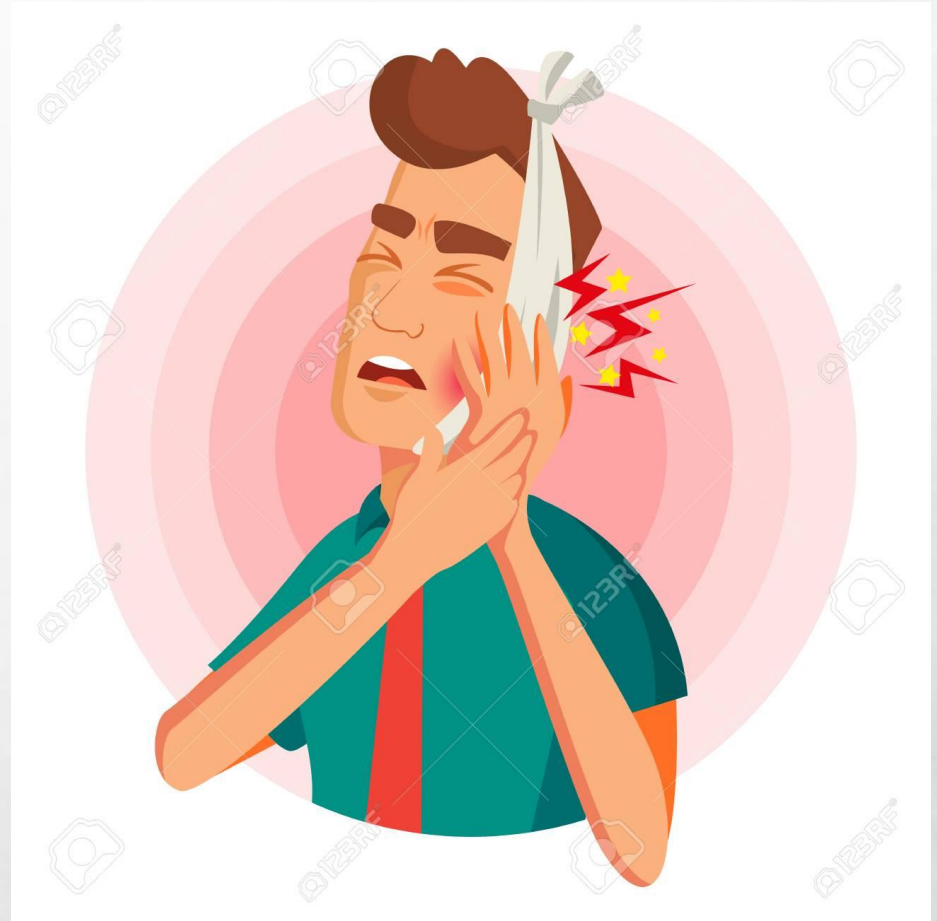


Sickle

R

+

Rake





+



Shovel



+



Hand fork



+



spade

QUESTION...

Imagine you are helping your family prepare a small farm for planting vegetables. You see different farm machines and tools in the shed: a plow, a harrow, a shovel, and even a small hand tractor. You are very eager to use them because they look powerful and exciting. However, you notice that some tools have sharp edges, some are heavy, and others make loud noises when turned on. If you were asked to choose one implement or piece of equipment to use first, which one would you choose and why? What possible dangers or accidents might happen if you don't know how to use it properly?

FARM IMPLEMENTS

Farm implements are larger, more complex devices or machines used in farming. They are often attached to or pulled by animals or tractors and are designed for heavier, large –scale tasks such as plowing, harrowing, planting and harvesting.

SOIL CULTIVATION

Farm Implements

Definition

Uses

Maintenance



Plow – used for breaking up tough soil, creating furrows for planting and burying residue.

Initial land preparation, incorporating amendments and burying crop residue.

Regular lubrication of moving parts, checking for worn parts like plowshares and proper storage to prevent rust.

Farm Implements

Definition

Harrow – break up clods, levels the soil surface and distributes residue evenly.

Uses

After plowing to refine soil texture: incorporating amendments like fertilizers.

Maintenance

Inspect for loose tines and replace if needed. Clean after use to prevent rust.



Farm Implements

Definition

Uses

Maintenance



Disc Harrow – uses rotating discs to cut and break up soil more aggressively than traditional harrow.

Often used for hard or compacted soil; can also be used for residue management.

Regular lubrication, inspect for worn or bent discs, and proper storage.

Farm Implements	Definition	Uses	Maintenance
	<u>Cultivator</u>– loosens soil between crop rows for weed control and aeration.	Used throughout the growing season to control weeds and improve air circulation around plants.	Regular lubrication, inspect for worn or bent discs, and proper storage.

PLANTING

Farm Implements

Definition

Seed Drill—precisely plants seeds at a specific depth and spacing.



Uses

Efficient planting of various seeds; often used for row crops like corn and soybeans.

Maintenance

Clean regularly to remove debris buildup, calibrate seeding mechanism periodically, and store indoors when not in use.

Farm Implements

Definition

Transplanter – mechanically transplants seedlings into the field.

Uses

Faster transplanting of seedlings compared to manual methods; suitable for large operations.

Maintenance

Regularly clean and lubricate, ensure proper adjustment for different seedling sizes and store indoors.





HARVESTING

Farm Implements	Definition	Uses	Maintenance
	<u>Combine Harvester</u> – cuts, threshes and separates grain from the stalks in a single pass.	Efficient harvesting of grains like wheat, barley, corn, and some types of beans.	It requires specialized service and maintenance due to complexity. Ensure proper storage of the equipment after harvest.



Farm Implements	Definition	Uses	Maintenance
	<u>Mower</u>– cuts forage crops like hay or alfalfa	Preparing animal feed or maintaining pastures	Sharpen blades regularly, lubricate moving parts and store in a dry place.



Farm Implements	Definition	Uses	Maintenance
	<u>Potato Harvester</u> – mechanically digs and collects potatoes.	Preparing animal feed or maintaining pastures	Sharpen blades regularly, lubricate moving parts and store in a dry place.



FARM EQUIPMENT

Farm equipment refers to motorized vehicles and machinery used for large-scale farming operations. they are very expensive and require significant technical knowledge to operate and maintain.

Farm Equipment

Definition

Types

Uses

Maintenance



Tractor – it is a powerful motorized vehicle with large wheels or tracks used for pulling or overpowering farm implements and equipment. It is the workhorse of any farm, providing power for various attachments.

Wheeled tractors (general purpose), track tractors (better for soft ground), compact tractors (smaller and more maneuverable)

Pulling trailers, powering implements, operating mowers and tilling soil.

Regular oil changes, filter replacements, checking tire pressure and fluid levels, inspecting for leaks and loose parts.

TILLAGE EQUIPMENT

- **Used for preparing soil for planting.**

Farm Equipment

Definition



Plows - various types (moldboard, chisel, disc) for breaking tough soil, creating furrows and burying crop residue.

Farm Equipment

Definition



Harrows - break up clods, level soil and distribute residue

Farm Equipment

Definition



Cultivators – loosen soil between crop rows for weed control and aeration.

PLANTING EQUIPMENT

- **Used for precisely placing seeds or seedlings in the ground.**

Farm Equipment	Uses	Farm Equipment	Uses	Farm Equipment	Uses
	<u>Seed drills</u> – plant seeds at a specific depth and spacing for row crops		<u>Transplanters</u> – mechanically transplant seedlings into the field; ideal for large operations.		<u>Broadcast seeders</u> – spread seeds over a larger area; suitable for cover crops or certain vegetables.



- ***USES***

- **Efficiently spreading seeds for crops like clover or some vegetable.**

- ***MAINTENANCE***

- **Regularly clean to remove debris buildup, calibrate settings for different seed sizes and store indoors.**

Farm Equipment	Uses	Farm Equipment	Uses	Farm Equipment	Uses	Farm Equipment	Uses
	<u>Combine harvester</u> <u>S</u> – cuts, threshes and separates grain from stalks in a single pass.		<u>Mowers</u> – cuts forage crops like hay and alfalfa		<u>Potato Harvester</u> – mechanically digs and collects potatoes.		<u>Fruit harvester</u> – shaking machines or conveyor belts for specific fruits like apples and grapes

- ***USES***

- **Efficient harvesting of fruits that can be mechanically detached from trees or vines.**

- ***MAINTENANCE***

- **Regularly lubricate, inspect for loose parts and wear and store properly after harvest.**

IRRIGATION EQUIPMENT

- **Provides water to crops during dry periods.**

Farm Equipment

Definition

Center Pivot Irrigation System – rotate a sprinkler on a central pivot point to water a circular area

Uses

Efficient watering of large fields to reduce labor costs.

Maintenance

Regularly inspect for leaks and clogs; ensure proper sprinkler function.



Farm Equipment	Definition	Uses	Maintenance
	<u>Drip Irrigation System</u> – slowly deliver water directly to the plant roots through a network of tubes.	Conserve water; ideal for high-value crops or areas with limited water resources.	Check for clogged emitters, ensure proper pressure and clean filters regularly.

AGRICULTURAL HAZARDS AND RISKS IN FARM OPERATIONS

- **Being a farmer is a pretty cool job – you get to work outdoors, grow food, be around animals. But just like any other job, there can be hazards and risks involved. A hazard is any potential source of harm, while risk refers to the likelihood of that harm occurring.**

CATEGORIES OF COMMON HAZARDS IN THE FARM INDUSTRY

1. PHYSICAL HAZARDS

- a. Slips, trips and falls:** uneven ground, wet surfaces and tools left around can cause nasty tumbles.
- b. Machinery:** tractors, harvesters and other equipment can be dangerous if not used properly.
- c. Falling objects:** watch out for branches, tools or anything else that could come crashing down.
- d. Noise and vibration:** farm equipment can be loud and cause hearing damage or muscle fatigue.

2. CHEMICAL HAZARDS

a. Pesticides and herbicides: these can be poisonous if inhaled or absorb through the skin.

b. Fertilizers: dust from fertilizers can irritate your lungs.

C. Fuel and lubricants: these can be flammable and cause skin irritation.

3. BIOLOGICAL HAZARDS

a. Animal bites and kicks: working with animals can be risky, especially if they are scared or stressed.

b. Zoonotic diseases: these are diseases that can be spread from animals to humans.

c. Mold and dust: exposure to mold and dust can cause respiratory problems.

4. ENVIRONMENTAL HAZARDS

- a. Sun exposure:** working long hours outdoors can lead to sunburn and heatstroke.
- b. Lightning:** be aware of thunderstorms and seek shelter if necessary.
- c. Floods and droughts:** these can damage crops and livestock.

MINIMIZING RISKS

- 1. Always wear proper Personal Protective Equipment (PPE), such as clothing and safety gear, like gloves, boots, goggles and hats.**
- 2. Never operate machinery without proper training.**
- 3. Be aware of your surroundings and watch out for potential hazards.**

MINIMIZING RISKS

- 4. Be cautious when handling chemicals, and follow the instructions on the label carefully.**
- 5. Wash your hands thoroughly after handling animals, chemicals or anything else that could be contaminated.**
- 6. Stay hydrated and take breaks to avoid heat exhaustion.**

QUESTIONS.....

- **WHAT IS THE MAIN USE OF A HAND TRACTOR IN CROP PRODUCTION?**
- **WHY IS IT IMPORTANT TO USE A PLOW BEFORE PLANTING CROPS?**
- **A FARMER USES A HAND TRACTOR WITHOUT WEARING PROTECTIVE GEAR. WHAT POSSIBLE HAZARDS OR ACCIDENTS MIGHT HAPPEN? HOW COULD THESE HAVE BEEN PREVENTED?**
- **COMPARE THE USE OF TRADITIONAL TOOLS (LIKE A PLOW PULLED BY ANIMALS) AND MODERN EQUIPMENT (LIKE A TRACTOR). WHICH DO YOU THINK IS MORE BENEFICIAL FOR SMALL-SCALE FARMERS? SUPPORT YOUR ANSWER WITH REASONS.**

LEARNERS' TAKEAWAYS

- **AFTER LEARNING ABOUT DIFFERENT FARM IMPLEMENTS AND EQUIPMENT, AS WELL AS THE HAZARDS AND RISKS IN USING THEM, WHAT THREE IMPORTANT LESSONS WILL YOU ALWAYS REMEMBER AND APPLY IF YOU ARE ASKED TO WORK IN A FARM OR SCHOOL GARDEN, AND WHY?**

REFLECTION ON LEARNING

- **AFTER STUDYING FARM IMPLEMENTS, EQUIPMENT, AND THE POSSIBLE HAZARDS AND RISKS IN USING THEM, HOW HAS YOUR UNDERSTANDING OF SAFETY IN FARMING CHANGED? IN WHAT WAYS CAN YOU APPLY THIS KNOWLEDGE TO PROTECT YOURSELF AND OTHERS WHEN WORKING IN A GARDEN OR FARM SETTING?**

FORMATIVE ASSESSMENT

1. WHICH FARM IMPLEMENT IS USED TO BREAK AND INVERT THE SOIL IN PREPARATION FOR PLANTING?

A. HARROW

B. PLOW

C. CULTIVATOR

D. SEEDER

2. A FARMER WANTS TO PLANT RICE SEEDLINGS QUICKLY AND UNIFORMLY IN THE FIELD. WHICH IMPLEMENT SHOULD HE USE?

A. SEED DRILL

C. CULTIVATOR

B. TRANSPLANTER

D. MOWER

FORMATIVE ASSESSMENT

3. WHICH FARM EQUIPMENT IS CONSIDERED THE MOST VERSATILE BECAUSE IT CAN PULL PLOWS, HARROWS, CULTIVATORS, AND EVEN OPERATE HARVESTERS?

A. SEEDER

B. TRACTOR

C. IRRIGATION SYSTEM

D. COMBINE HARVESTER

FORMATIVE ASSESSMENT

4. A FARMER IS APPLYING PESTICIDES IN THE FIELD. WHICH TYPE OF AGRICULTURAL HAZARD IS HE MOST EXPOSED TO?

A. PHYSICAL HAZARD

B. CHEMICAL HAZARD

C. BIOLOGICAL HAZARD

D. ENVIRONMENTAL HAZARD

5. WORKING UNDER EXTREME HEAT IN THE FIELD MAY CAUSE DEHYDRATION OR HEATSTROKE. THIS IS AN EXAMPLE OF WHICH HAZARD?

A. BIOLOGICAL HAZARD

B. CHEMICAL HAZARD

C. ENVIRONMENTAL HAZARD

D. PHYSICAL HAZARD

ANSWER KEY

1 – B

2 – B

3 – B

4 – B

5 – C

ACTIVITY

- 1. THE CLASS WILL BE DIVIDED INTO 8 GROUPS.**
- 2. THEN THE TEACHER WILL GIVE THE ACTIVITY TO EACH GROUP.**